



BAULKHAM HILLS HIGH SCHOOL

2021

HSC Assessment Task 1

Mathematics Extension 2

General Instructions

- Reading time – 5 minutes
- Working time – 60 minutes
- Write using non-erasable black or blue pen.
- Calculators approved by NESA may be used.
- Marks may be deducted for careless or badly arranged work.
- Relevant formulae from the NESA Reference Sheet are provided on page 2.
- Answer all questions on the appropriate page of your answer booklet.
- For questions in Section II, show relevant mathematical reasoning and/or calculations.
- If extra paper is used, it must be clearly labelled with your name and the question being answered, and stapled inside your answer booklet.

Total marks – 43

Exam consists of 5 pages.

Section I: Pages 2-4

5 marks

Attempt Questions 1-5

Section II: Pages 5-7

38 marks

Attempt Questions 6-9

Complex Numbers

$$z = a + ib = r(\cos \theta + i \sin \theta) \\ = re^{i\theta}$$

$$\left[r(\cos \theta + i \sin \theta) \right]^n = r^n(\cos n\theta + i \sin n\theta) \\ = r^n e^{in\theta}$$

Section I:

Marks

5 marks

Attempt Questions 1-5

Use the multiple choice answer sheet in your booklet.

1. If $z = \sqrt{3} + i$, what is the value of $\overline{\left(\frac{i}{z}\right)}$?

1

(A) $\frac{1+i\sqrt{3}}{4}$

(B) $\frac{-\sqrt{3}+i}{4}$

(C) $\frac{1-i\sqrt{3}}{4}$

(D) $\frac{\sqrt{3}-i}{4}$

2. ω is a non-real root of the equation $z^5 + 1 = 0$.

1

Which of the following is not a root of this equation?

(A) $\bar{\omega}$

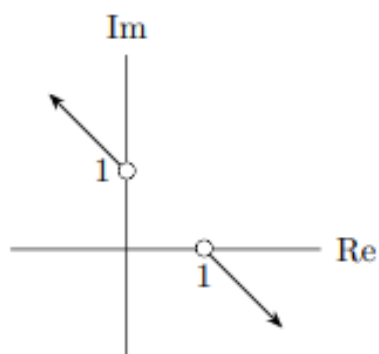
(B) ω^2

(C) $\frac{1}{\omega}$

(D) ω^3

3. The locus of a complex number z is shown on the Argand diagram below.

1



Which of the following is the equation of this locus?

- (A) $\arg(z - i) - \arg(z - 1) = 0$
- (B) $\arg(z - i) - \arg(z - 1) = \pi$
- (C) $\arg(z + i) - \arg(z + 1) = 0$
- (D) $\arg(z + i) - \arg(z + 1) = \pi$
4. What is the smallest positive value of θ such that $e^{i\theta} \times e^{2i\theta} = i$?

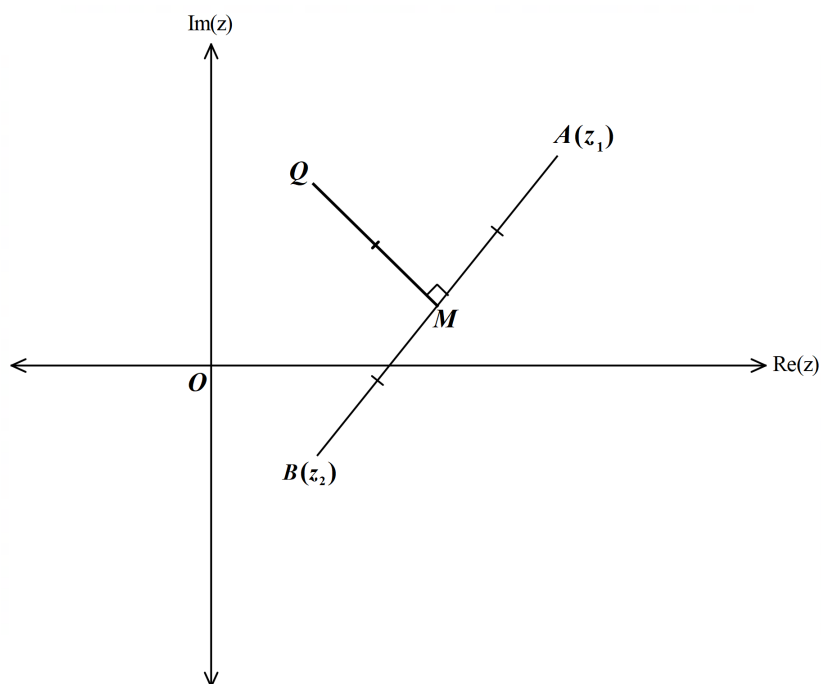
1

- (A) $\frac{\pi}{12}$
- (B) $\frac{\pi}{6}$
- (C) $\frac{\pi}{3}$
- (D) $\frac{5\pi}{6}$

5. The complex numbers z_1 and z_2 are represented by points A and B respectively.

1

QM is perpendicular to AB, and $QM = AM = BM$.



What is the complex number that corresponds to Q ?

- (A) $i\left(\frac{z_1 - z_2}{2}\right)$
- (B) $i\left(\frac{z_1 + z_2}{2}\right)$
- (C) $\left(\frac{z_1 + z_2}{2}\right) + i\left(\frac{z_1 + z_2}{2}\right)$
- (D) $\left(\frac{z_1 + z_2}{2}\right) + i\left(\frac{z_1 - z_2}{2}\right)$

End of Section I

Section II: 38 marks

Attempt Questions 6-9

Question 6 (11 marks)

a) Simplify

(i) $|2 + 3i|$ 1

(ii) $\text{Arg}(-1 - i\sqrt{3})$ 2

(iii) $\frac{4 + 3i}{1 - 2i}$ 2

b) Find the square roots of $9e^{i\frac{\pi}{5}}$ expressing your answers in simplest exponential form. 2

c) Given $|z - 2 - 2i| = 1$:

(i) Draw the locus of z on the Argand diagram. 2

(ii) Find the maximum value of $|z|$ 1

(iii) Find the maximum value of $\text{Arg}(z)$, giving your answer in radians correct to 3 decimal places. 1

Question 7 (9 marks)

- a) (i) Find the two square roots of $15 - 8i$, giving the answers in the form $x + iy$, where x and y are real numbers. 2
- (ii) Hence, or otherwise, solve the equation $z^2 - (2 + 3i)z - 5 + 5i = 0$ giving your solutions in the form $a + ib$ where a and b are real numbers. 3
- b) The polynomial $P(z) = 3z^4 - z^3 + 3z^2 + 29z - 10$ has a root $z_1 = 1 - 2i$.
- (i) Explain why $1 + 2i$ is also a root. 1
- (ii) Hence express $P(z)$ as a product of real linear and real quadratic factors. 3

Question 8 (10 marks)

- a) z is a complex number which satisfies $\arg(z + 2) - \arg(z - 2) = \frac{\pi}{4}$.
- (i) Sketch the locus of z . 2
- (ii) Find the Cartesian equation of this locus, noting any restrictions. 2
- b) (i) By using De Moivre's theorem and the binomial expansion of $(\cos \theta + i \sin \theta)^5$, show that $\cos 5\theta = 16 \cos^5 \theta - 20 \cos^3 \theta + 5 \cos \theta$. 3
- (ii) Hence solve the equation $\cos 5\theta = \cos \theta$ for $0 \leq \theta \leq 2\pi$. 3

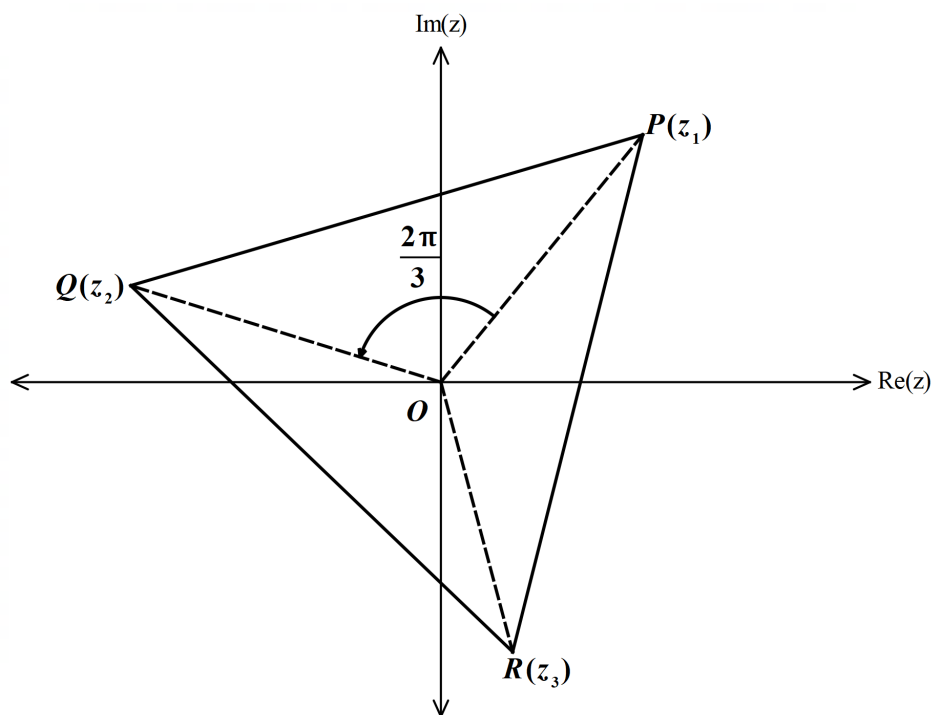
Question 9 (8 marks)

a) Given $1 - i$ is a root of $P(z) = z^3 - (1 + 2i)z^2 + (1 + 3i)z + 2 - 2i$, find the other roots. **3**

b) The points P , Q and R on the Argand diagram represent the complex numbers z_1 , z_2 and z_3

so that PQR is an equilateral triangle whose centre is the origin (ie. $OP = OQ = OR$) and

$$\angle POQ = \frac{2\pi}{3}.$$



Let $\omega = \cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3}$.

(i) Show that $1 + \omega + \omega^2 = 0$. **1**

(ii) Find an expression for z_2 in terms of z_1 and ω . **1**

(iii) The complex number z_4 is represented by point S such that $PQRS$ is a parallelogram. **3**

Using vector methods, express z_4 in terms of z_1 and ω giving your answer in its **simplest** form.

END OF EXAMINATION

Section I.

$$1. \quad \overline{\left(\frac{i}{z}\right)} = \frac{-i}{\sqrt{3}-i} \cdot \frac{\sqrt{3}+i}{\sqrt{3}+i}$$

$$= \frac{1-i\sqrt{3}}{4} \quad (C)$$

$$2. \quad \omega^5 = -1 \quad \text{since } \omega \text{ is a root.}$$

$$\text{If } z = \omega^2, \quad z^5 + 1 = (\omega^2)^5 + 1$$

$$= (\omega^5)^2 + 1$$

$$= (-1)^2 + 1$$

$$\neq 0 \quad (B)$$

3. (A)

$$4. \quad e^{i\theta} \times e^{2i\theta} = i$$

$$e^{3i\theta} = e^{i\pi/2}$$

$$3\theta = \frac{\pi}{2} + 2k\pi \quad (k \in \mathbb{Z})$$

$$\theta = \frac{\pi}{6} \quad (B)$$

5. Q and $\vec{OQ}$ represent the same complex number.

$$\vec{OQ} = \vec{OM} + \vec{MQ}$$

$$= \frac{z_1 + z_2}{2} + i \left(\frac{z_1 - z_2}{2} \right) \quad (D)$$

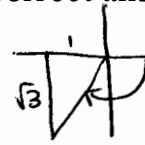
SECTION II

Question 6.

a) (i) $\sqrt{2^2 + 3^2} = \sqrt{13}$

(1) correct answer

(ii) $\text{Arg}(-1 + i\sqrt{3}) = \text{Arg}(-1 - i\sqrt{3})$
 $= -\frac{2\pi}{3}$

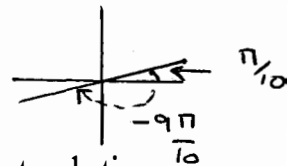


(2) correct soln
 (1) obtains $\pi/3$

(iii) $\frac{4+3i}{1-2i} \cdot \frac{1+2i}{1+2i} = \frac{4-6+11i}{5}$
 $= -\frac{2}{5} + \frac{11}{5}i$
 $\therefore \text{Re}\left(\frac{4+3i}{1-2i}\right) = -\frac{2}{5}$

(2) correct answer
 (1) multiply by conjugate

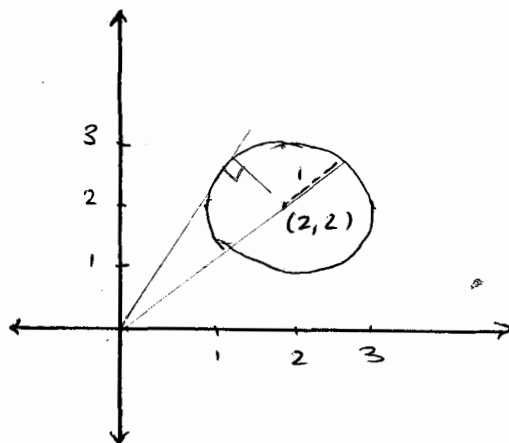
b) $\sqrt{9e^{i\pi/5}} = \pm 3e^{i\pi/10}$
 $= 3e^{i\pi/10}, 3e^{i(-9\pi/10)}$



(2) correct solution
 (1) one correct root

c) $|z - 2 - 2i| = 1$

(i)



← Locus is a circle
 centre (2, 2), radius 1 unit.

(2) correct locus
 (1) circle with correct centre or radius

(ii) $\text{Max } |z| = \sqrt{8} + 1$
 $= 2\sqrt{2} + 1$

(1) correct answer

(iii) $\text{Max arg } z = \frac{\pi}{4} + \sin^{-1}\left(\frac{1}{\sqrt{8}}\right)$
 $= 1.147$

(1) correct answer

Question 7.

a) i) Let $z = x + iy$: $(x + iy)^2 = 15 - 8i$ $(x, y \in \mathbb{R})$

$$x^2 - y^2 + 2ixy = 15 - 8i$$

$$x^2 - y^2 = 15 \quad \text{--- ①}$$

$$2xy = -8$$

$$xy = -4$$

$$y = \frac{-4}{x} \quad \text{--- ②}$$

$$x^2 - \frac{16}{x^2} = 15$$

$$x^4 - 15x^2 - 16 = 0$$

$$(x^2 - 16)(x^2 + 1) = 0$$

$$x^2 = 16$$

$$\cancel{x^2 = -1}$$

$$x = \pm 4$$

$$(x \in \mathbb{R})$$

$$y = \mp 1$$

$$\therefore z = 4 - i, -4 + i$$

$$\text{i.e. } z = \pm (4 - i)$$

(2) correct soln

(1) obtains simultaneous eqns and attempts to solve

(ii) $z^2 - (2 + 3i)z - (5 - 5i) = 0$

$$\Delta = [-(2 + 3i)]^2 - 4 \cdot 1 \cdot -(5 - 5i)$$

$$= 4 - 9 + 12i + 20 - 20i$$

$$= 15 - 8i$$

$$z = \frac{2 + 3i \pm (4 - i)}{2}$$

$$= \frac{6 + 2i}{2} \quad , \quad \frac{-2 + 4i}{2}$$

$$= 3 + i, -1 + 2i$$

(3) correct soln

(2) attempts to link result from (i) to quadratic formula

(1) finds discriminant

Question 7 (cont)

- b) (i) Since $1-2i$ is a root, and all the coefficients are real, its conjugate must also be a root

$$(1+2i)$$

(1) correct explanation

- (ii) There is a quadratic factor: $(S=2, P=5)$

$$z^2 - 2z + 5$$

$$\therefore 3z^4 - z^3 + 3z^2 + 29z - 10$$

$$= (z^2 - 2z + 5)(3z^2 + 5z - 2)$$

$$= \underline{(z^2 - 2z + 5)(3z - 1)(z + 2)}$$

15
-2

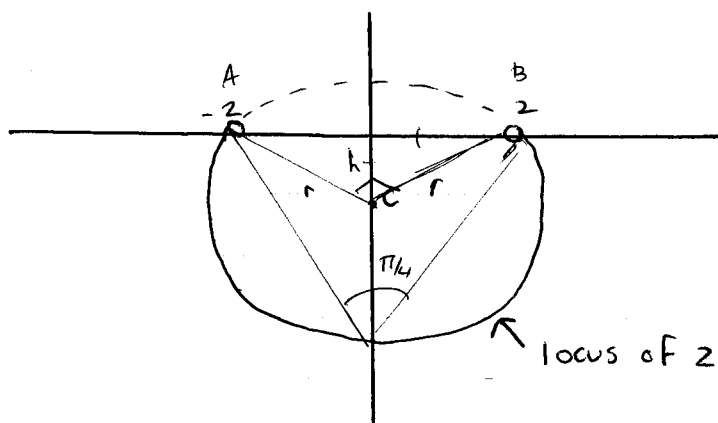
(3) correct soln

(2) finds a second quadratic factor

(1) finds one quadratic factor

Question 8

a) (i)



- (2) correct arc, endpoints excluded
- (1) a major arc with correct endpoints

(ii) $2r^2 = 4^2$
 $r^2 = 8$
 $r = 2\sqrt{2}$
 $h = 2$
 $\therefore \text{Centre } (0, -2)$

~~Cent~~

$\therefore \text{Eqn: } x^2 + (y+2)^2 = 8$
 with $y < 0$

- (2) correct eqn and restriction
- (1) correct eqn

b) (i) $(\cos \theta + i \sin \theta)^5 = \cos 5\theta + i \sin 5\theta$ (by De Moivre's theorem)

$$(\cos \theta + i \sin \theta)^5 = c^5 + 5c^4is + 10c^3(is)^2 + 10c^2(is)^3 + 5c(is)^4 + (is)^5$$

$$= (c^5 - 10c^3s^2 + 5cs^4) + i(5c^4s - 10c^2s^3 + s^5)$$

where $c = \cos \theta, s = \sin \theta$

Equating the real parts:

$$\begin{aligned} \cos 5\theta &= \cos^5 \theta - 10\cos^3 \theta \sin^2 \theta + 5\cos \theta \sin^4 \theta \\ &= \cos^5 \theta - 10\cos^3 \theta (1 - \cos^2 \theta) + 5\cos \theta (1 - 2\cos^2 \theta + \cos^4 \theta) \\ &= \cos^5 \theta - 10\cos^3 \theta + 10\cos^5 \theta + 5\cos \theta - 10\cos^3 \theta + 5\cos^5 \theta \\ &= 16\cos^5 \theta - 20\cos^3 \theta + 5\cos \theta \end{aligned}$$

- (3) correct soln
- (2) 2 correct expansions and attempt to equate real parts
- (1) 2 correct expansions

Question 8 (cont).

b) (ii) $\cos 5\theta - \cos \theta = 0 \quad [0 \leq \theta \leq 2\pi]$

$$16\cos^5\theta - 20\cos^3\theta + 4\cos\theta = 0$$

$$4\cos\theta (4\cos^4\theta - 5\cos^2\theta + 1) = 0$$

$$4\cos\theta (4\cos^2\theta - 1)(\cos^2\theta - 1) = 0$$

$$\cos\theta = 0$$

$$\cos\theta = \pm \frac{1}{2}$$

$$\cos\theta = \pm 1$$

$$\therefore \theta = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{\pi}{3}, \frac{2\pi}{3}, \frac{4\pi}{3}, \frac{5\pi}{3}, 0, \pi, 2\pi$$

(3) All correct solutions

(2) Solves 2 of the three cases correctly

(1) Factorises LHS of the eqn

Question 9.

a) Let roots be $\alpha, \beta, 1-i$

$$\alpha + \beta + 1-i = 1+2i$$

$$\alpha + \beta = 3i \quad \text{--- (1)}$$

$$\alpha\beta(1-i) = -2+2i$$

$$\alpha\beta = \frac{-2+2i}{1-i} = \frac{-2(1-i)}{1-i} = -2 \quad \text{--- (2)}$$

$$\alpha(3i - \alpha) = -2$$

$$3i\alpha - \alpha^2 = -2$$

$$\alpha^2 - 3i\alpha - 2 = 0$$

$$(\alpha - 2i)(\alpha - i) = 0$$

$\therefore$ The other two roots are $2i, i$.

(Note: if $\alpha = 2i, \beta = i$ and if $\alpha = i, \beta = 2i$).

(3) correct soln.

(2) Solves simultaneous eqns arising from sum and product

(1) Forms eqns using sum and product of roots.

$$\begin{aligned} b) \quad 1 + \omega + \omega^2 &= 1 + \cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3} + \cos \frac{4\pi}{3} + i \sin \frac{4\pi}{3} \\ &\quad \text{(by De Moivre's)} \\ &= 1 + \left(-\frac{1}{2} + i \frac{\sqrt{3}}{2}\right) + \left(-\frac{1}{2} - i \frac{\sqrt{3}}{2}\right) \\ &= 1 - \frac{1}{2} + i \frac{\sqrt{3}}{2} - \frac{1}{2} - i \frac{\sqrt{3}}{2} \\ &= 0. \end{aligned}$$

OR

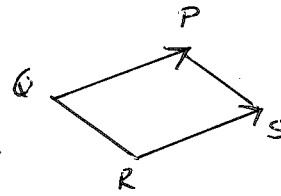
$$\begin{aligned} 1 + \omega + \omega^2 &= 1 + \text{cis } \frac{2\pi}{3} + \text{cis } -\frac{2\pi}{3} \\ &= 1 + 2 \cos \frac{2\pi}{3} \\ &= 1 + 2 \left(-\frac{1}{2}\right) \\ &= 0. \end{aligned}$$

(1) correctly shown

$$(ii) \quad z_2 = z, \omega$$

(1) correct answer

(iii) $\vec{QP} = \vec{RS}$ since
opposite sides of parallelogram
are parallel & equal.



$$z_1 - z_2 = z_4 - z_3$$

$$z_4 = z_1 - z_2 + z_3$$

$$= z_1 - z_1\omega + z_1\omega^2$$

$$= z_1(1 - \omega + \omega^2)$$

From (i), $1 + \omega + \omega^2 = 0$

$$\omega^2 = -1 - \omega$$

$$\therefore z_4 = z_1(1 - \omega - 1 - \omega) = \underline{-2\omega z_1}$$

(3) Correct soln

(2) Uses vectors to obtain z_4 in terms of z_1 and ω with reason

(1) Uses vectors to obtain a relationship between z_1, z_2, z_3, z_4